

# MORGAN OLIVIA EVANS

Orange County, California

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## PROFILE

Recent computer science graduate with hands-on experience in embedded systems, high-performance computing and scientific research. Experienced in leading technical teams, collaborating with educators and industry leaders, and managing end-to-end development projects. Seeking full-time roles that support continued learning and real-world impact, especially in earth system science and computing at scale.

## EDUCATION

**UNIVERSITY OF CALIFORNIA, IRVINE** – B.S. Computer Science, Cumulative GPA: 3.55

August 2023 – June 2025

**SANTA MONICA COLLEGE** – A.S. Computer Science, Transfer GPA: 3.85

June 2020 – June 2023

## TECHNICAL SKILLS

**Programming Languages** – Bash | C | C++ | Java | Python | CUDA

**Tools & Environments** – Linux CLI | Slurm | PBS | AWS | Azure | Arduino | ESP32 | UART | I2C | SPI | Git | Docker | CI/CD | Confluence | Jira

## PROFESSIONAL EXPERIENCE

**NASA Center for Climate Simulation** | High-Performance Computing Intern

June 2024 – August 2024

- Investigated MPI and I/O performance anomalies for the GEOS climate model on NCCS Discover (Goddard) and NAS Aitken (Ames) supercomputing clusters using Slurm and PBS job schedulers, as well as Azure and AWS cloud-based environments managed with Spack.
- GEOS performance benchmarking was conducted in two parts: OSU Micro-benchmarks on AMD Milan architecture across various platforms and I/O benchmarking on the NCCS Discover cluster to compare file and operating systems for performance impacts.
- Identified several scalability and performance issues, including increased latency with higher tasks per node.
- Directly contributed to advancing the performance of climate simulations run on NASA's supercomputing infrastructure.

**ACCESS** | Computational Research Intern

July 2023 – June 2024

- Used NVIDIA Nsight and Compute profiling tools and NVTX annotations to analyze GPU-accelerated ice-sheet flow modeling software.
- Reduced runtime by 10% by removing redundancies, establishing concurrency and eliminating expensive operations within CUDA kernels.
- Utilized two Slurm-managed supercomputing clusters to compare runtime, grid points, scale and error values between test cases.
- Research findings narrow the uncertainty in sea level rise prediction to better understand and limit the effects of climate change.

**Statewide California Earthquake Center** | Computational Research Intern

May 2022 – February 2023

- Developed Bash scripts to build computational seismic models with SW4 2.01 and visualized results using JRG-Viewmat software.
- Utilized the Rockfish supercomputing cluster at Johns Hopkins University and conducted an in-depth SW4 cluster scaling analysis and performance evaluation using Slurm.
- Research findings improve seismic model accuracy, refine hazard estimates and progress preparedness efforts.

## PROJECTS

**Visualization Tools for Structural Analysis** | UCI 2025 ICS Project Expo | 3<sup>rd</sup> Place Award

January 2025 – June 2025

- As part of UCI's computer science senior capstone program, I led a team of six students on a multi-quarter project for our sponsor, KPFF, a global structural engineering firm. KPFF tasked our team to develop a full-stack visualization software. We built the application using Godot Engine and GDScript, handling both backend OpenSees (Open System for Earthquake Engineering Simulation) TCL script processing and frontend 3D visualization of structural models and post-processing results.

**Embedded Audio Feedback Device**

May 2025 – June 2025

- Designed and built an embedded system using an ATmega32, discrete LEDs, and a sound decibel detector to visualize voice volume while gaming. Programmed state and volume thresholds in C to trigger LED indicators for low, moderate, elevated and excessive volume.

**CON-E** | BitHacks 2025 | Best Meme Hack Award

April 2025

- Designed and implemented the embedded system architecture for a Raspberry Pi-ESP32 autonomous robotic traffic cone. Integrated vision-based tracking, motor control, audio feedback, and RGB LED with custom 3D-printed chassis, internal components and power.

**Motion-Activated Motorized Cat Toy**

December 2024

- Designed and assembled an embedded circuit using an ESP32 with motor control, speaker, LED and accelerometer components. Programmed motion-activated behavior and autonomous state logic for interactive play. Created a 3D-printed spherical housing with interchangeable attachments and integrated internal components.